F. Minimal architectural requirements

To verify our theoretical understanding of the transformer's computational requirements, we conduct a systematic evaluation across different architectural configurations. Fig. A7 shows that the model achieves good performance with minimal architecture: a single layer with two attention heads is sufficient to achieve low KL divergence across different Mess3 parameter settings. This empirical finding aligns with our theoretical analysis - when $x > 1/3$, the belief update patterns contain oscillatory components that require two heads to implement due to the non-negativity constraint of attention. The necessity of two heads is visually demonstrated in Fig. A8. For $x = 0.5$, where the optimal update pattern has significant oscillatory components, a single-head transformer fails to capture the correct belief geometry. With two heads, the model can properly implement these updates through complementary attention patterns, resulting in representations that closely match the ground truth geometry.

G. Dimensionality of Residual Stream Activations

Table 1: Cumulative explained variance ratios for PCA components of the residual stream activations at the intermediate position (after attention) and the final position (before unembedding). The table shows results for different settings of the Mess3 HMM parameters x and α.

		Intermediate				Final			
	x	0.15	0.15	0.5	0.5	0.15	0.15	0.5	0.5
component	α	0.2	0.6	0.6	0.2	0.2	0.6	0.6	0.2
0		0.5408	0.4648	0.4074	0.5268	0.9618	0.4947	0.4596	0.6503
1		0.8768	0.8894	0.8028	0.8519	0.9825	0.7681	0.7096	0.8592
2		0.9673	0.9859	0.8913	0.9173	0.9943	0.9811	0.8855	0.9689
3		0.9749	0.9903	0.9455	0.9649	0.9960	0.9897	0.9189	0.9755
4		0.9815	0.9929	0.9848	0.9886	0.9969	0.9916	0.9428	0.9807
5		0.9870	0.9942	0.9978	0.9977	0.9976	0.9931	0.9586	0.9850
6		0.9914	0.9955	0.9986	0.9984	0.9981	0.9945	0.9723	0.9886

We perform PCA on the residual stream activations after the attention module (intermediate) and before the unembedding layer (final). The effective dimensionality of the residual stream is low, with the first few components capturing most of the variance (See Table 1). In most cases, the first 3 components explain over 90% of the variance. For $x = 0.5$, the effective dimensionality is higher, possibly due to the oscillatory dynamics of the belief updating equation in this regime. Further investigation is needed to fully understand this phenomenon.

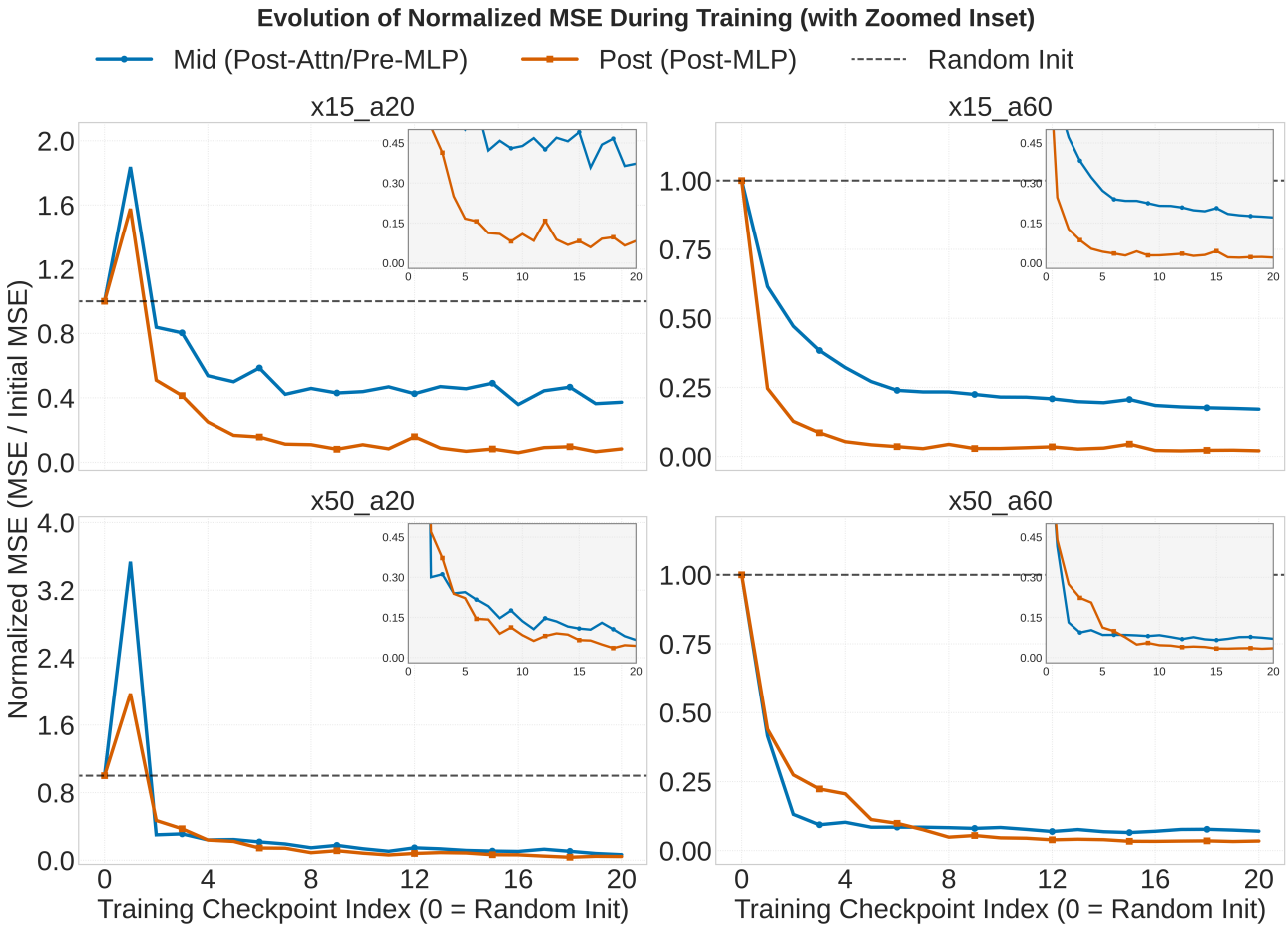

Figure A3: **Evolution of Normalized Mean Squared Error (MSE) During Training Across Experimental Conditions.** Each subplot shows the MSE for one experimental condition (labeled by title), normalized by the initial MSE value from the randomly initialized model (represented at Checkpoint Index 0). The Y-axis represents MSE relative to this random baseline (Lower is better), where the dashed line at Y=1 indicates performance equal to random initialization. The X-axis represents the training checkpoint index, starting from the random initialization at index 0. Lines: Show the normalized MSE for Mid (Post-Attention/Pre-MLP, blue circles) and Post (Post-MLP, orange squares) activations, reflecting their fit to their respective theoretical geometries over training. For Mid activations we regress to the Constrained Belief geometry, while for Post activations we regress to the Full (unconstrained) Belief geometry. Insets: Provide a zoomed-in view of the Y-axis from -0.02 to 0.5, highlighting the convergence behavior at low MSE values. **Observations:** Across all conditions, both Mid and Post activation representations show significantly improved geometric fits (normalized MSE drops well below 1) compared to the random baseline as training progresses. Some conditions (e.g., a=20) may show an initial transient increase in normalized MSE before rapid improvement.

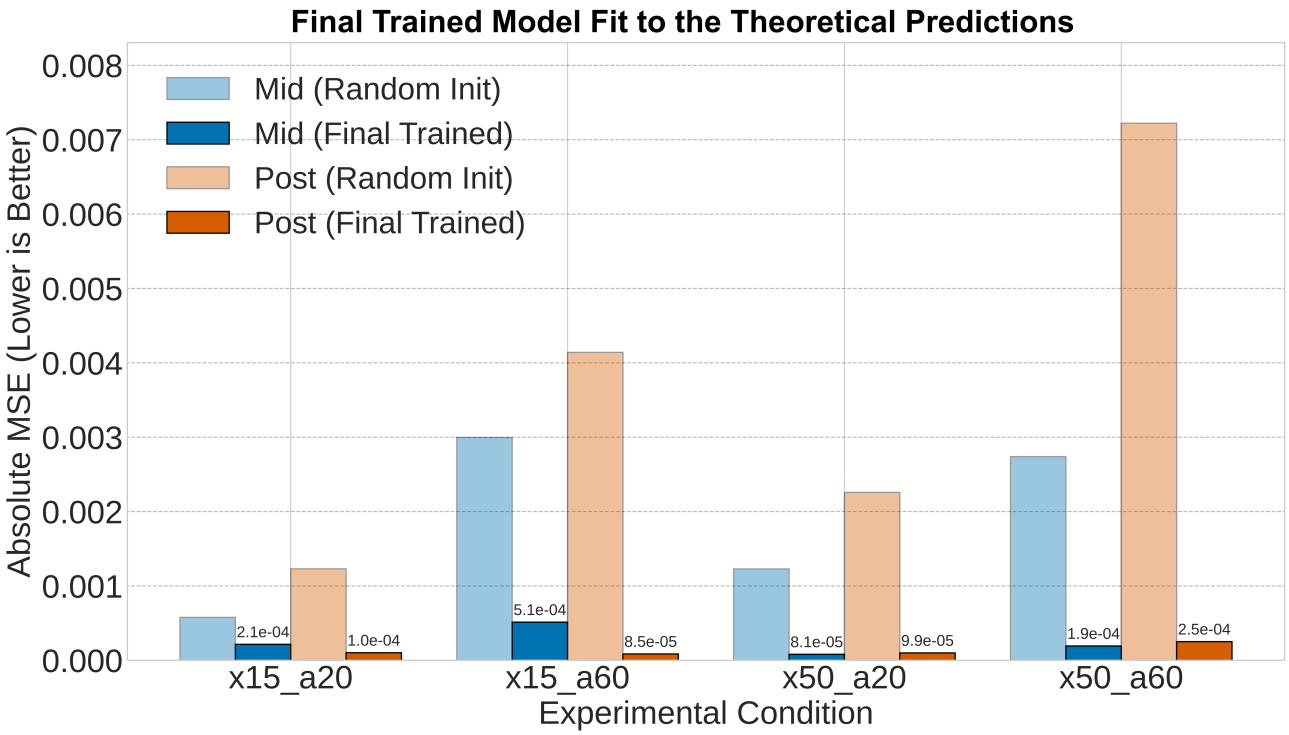

Figure A4: **Comparison of Final Trained Model Fit vs. Random Initialization Across Experimental Conditions.** The bar chart displays the Absolute Mean Squared Error (MSE, lower is better), comparing the geometric fit of activations at the final training checkpoint against the initial random model state. Results are shown for four different experimental conditions (X-axis). Within each condition, bars represent: Mid (Post-Attention/Pre-MLP) Activations, MSE of the regression to the Constrained Belief Geometry: Blue bars. Post (Post-MLP) Activations, MSE of the regression to the Full (unconstrained) Belief Geometry: Orange bars. Solid bars indicate the MSE for the final trained model, while transparent bars of the same color show the MSE for the randomly initialized model (baseline). Numerical labels specify the precise MSE values achieved by the trained models. **Observations:** Across all conditions, the trained model (solid bars) achieves lower MSE, indicating a much better fit to the underlying theoretical geometries compared to the random baseline (transparent bars).